

Risk, Return, Diversification, and Trend-following Strategies: Evidence from U.S. Technology and Kazakhstan- Related Equities

Independent Research Project

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Chapter 1: Introduction

Investment decisions involve a trade-off between the returns investors seek and the risks they are willing to bear, so comparing assets by return alone gives an incomplete picture. Volatility, drawdown, and risk-adjusted return show how much uncertainty and loss an investor endured to earn a given return.

Diversification is a second key consideration. Markowitz (1952) showed that combining assets whose returns are not perfectly correlated can reduce portfolio risk relative to holding each asset alone. The benefit, however, varies across asset groups and over time, so it is worth measuring in a specific sample rather than assuming it.

Investors also use trend-following strategies. A moving-average strategy compares an asset's price with its historical average and adjusts its position based on the resulting signal. Such rules are simple to implement, but their performance depends on the moving-average window, the period examined, and trading costs. Backtests that ignore transaction costs or compare strategies over inconsistent periods can overstate a strategy's usefulness (Ready, 2002).

This study examines six equities in two groups: Apple, Microsoft, and NVIDIA as U.S. technology stocks, and Kaspi.kz, Halyk Bank, and Kazatomprom as Kazakhstan-related stocks. The Kazakhstan-related group consists of companies associated with Kazakhstan that trade under different listing arrangements; it is not a representative sample of the Kazakhstan stock market. The study compares the two groups on return, risk, and diversification, and tests moving-average strategies against a buy-and-hold benchmark.

Portfolio comparison. Each group is held as an equal-weighted portfolio over a matched sample from 22 January 2024 to 31 December 2024 (235 observations per portfolio). Aligning the dates ensures both portfolios are judged over the same period. Performance is measured by annualized arithmetic return, annualized volatility, the Sharpe ratio, cumulative return, maximum drawdown, and diversification measures. The results describe these securities over this period only and should not be read as estimates for either country's equity market.

Strategy analysis. The study compares MA20, MA50, MA100, and MA200 rules with buy and hold. Signals are lagged by one trading day, so a signal formed from one day's closing price is never applied to that same day's return. Strategies are evaluated on return and risk measures under transaction costs of 0, 10, and 25 basis points. Comparisons begin only after each moving average has warmed up, and all strategies are judged over a common evaluation period. The analysis also examines annual outperformance and stock-year observations in which buy and hold lost money.

Research questions

1. How do the return and risk characteristics of the selected U.S. technology and Kazakhstan-related equities differ over the matched sample?

2. How do the correlations and diversification ratios of the two equal-weighted portfolios compare?
3. How do the MA20, MA50, MA100, and MA200 strategies perform relative to buy and hold?
4. How do transaction-cost assumptions affect strategy returns and risk-adjusted performance?

This study offers a focused comparison of selected equities and a transparent test of simple trend-following rules under stated assumptions. Its scope is limited by the small number of securities, the short matched sample, differences in listing venues and trading calendars, and the simplified backtest. These limitations are taken into account when interpreting the results.

Chapter 2: Literature Review

2.1 Risk and Return

Return is the gain or loss on an investment over a period, and risk is the uncertainty around that return. Empirical studies typically describe historical performance using realized returns and volatility. These measures are useful for comparison but do not guarantee future results (Bodie, Kane, & Marcus, 2021).

The risk-return trade-off suggests that assets with higher risk should be compensated with higher returns. This study only observes whether higher realized volatility coincided with higher realized returns in its sample. It does not estimate expected returns or test whether risk causes return.

Because two investments can differ in both return and volatility, risk-adjusted measures are also needed. The Sharpe ratio divides a portfolio's excess return over the risk-free rate by the volatility of its returns (Sharpe, 1966). This study uses the Sharpe ratio computed on excess returns, with the risk-free-rate assumption stated in the methodology. A simple ratio of mean return to volatility is not treated as a Sharpe ratio.

2.2 Modern Portfolio Theory and Portfolio Construction

Modern Portfolio Theory holds that an asset should be judged by how it contributes to a portfolio, not only by its own risk and return. Markowitz (1952) showed that portfolio risk depends on the risk of each asset and on the relationships among their returns. When assets are not perfectly correlated, combining them can reduce portfolio volatility.

Diversification does not guarantee higher returns or better risk-adjusted performance. Its benefit depends on the assets, their correlations, their weights, and the period examined. The theory therefore motivates comparing portfolio risk and correlations, but it does not imply that a portfolio will outperform its individual holdings.

This study uses equal-weighted portfolios, with one-third allocated to each of three securities, and compares a U.S. technology portfolio with a Kazakhstan-related portfolio. The weights are fixed rather than optimized using expected returns, a covariance matrix, an efficient frontier, or a maximum-Sharpe procedure. The analysis therefore compares two equal-weighted portfolios; it does not optimize portfolio construction.

2.3 Diversification and Correlation

Diversification is commonly assessed by how asset returns move together. Pairwise correlation measures the strength and direction of the relationship between two return series. Lower correlations offer more room for volatility reduction when assets are combined, although correlation alone does not capture every aspect of portfolio risk.

This study uses two correlation measures: full-sample pairwise correlations and 30-day rolling pairwise correlations, which allow the relationships among assets to change over time. It also calculates a diversification ratio, defined as the weighted average volatility of the individual assets divided by the portfolio's volatility. The ratio summarizes how much volatility the portfolio removes relative to its components (Choueifaty & Coignard, 2008).

Choueifaty and Coignard use the ratio to construct a maximum-diversification portfolio. This study does not perform that optimization; it computes the ratio for equal weights as a descriptive comparison of the two portfolios.

Diversification theory also distinguishes firm-specific risk from risk that affects many assets or the whole market. This study mentions the distinction as background but does not separate systematic from unsystematic risk. Its evidence is limited to correlations, portfolio volatility, and the diversification ratio for the selected securities.

2.4 Moving-Average Trend-Following Strategies

Moving averages smooth past prices over a selected window. A price-versus-moving-average strategy can use the relationship between the current price and its moving average to determine whether to hold an asset or remain in cash. Shorter windows respond more quickly to price changes, while longer windows respond more slowly. This study tests MA20, MA50, MA100, and MA200.

The strategy tested here is a **price-versus-single-moving-average rule**: it holds the asset when its price is above the selected moving average and otherwise holds no position. This is different from a moving-average crossover rule, which compares two moving averages with each other. The strategy signal is shifted by one trading day before being applied to returns to avoid using the same day's closing-price signal to capture that day's return.

Research on technical trading rules reports varied findings. Brock, Lakonishok, and LeBaron (1992) found evidence of predictive power for moving-average and trading-range rules in historical Dow Jones data. Bessembinder and Chan (1998) revisited technical-rule results and discussed how nonsynchronous trading and transaction costs affect their interpretation. Glabadanidis (2015) reported positive results for moving-average timing strategies across a much broader set of U.S. and international assets. These studies use different securities, periods, rules, and methods; their findings do not establish how the strategies will perform for the six securities in this study.

The discussion of market efficiency provides another reason to test rather than assume that a simple rule is profitable. Under the efficient-markets framework, security prices reflect available information, which makes persistent abnormal performance difficult to achieve (Fama, 1970). The empirical question is therefore whether the specified moving-average rules perform differently from buy and hold in the sample, not whether moving averages are universally effective.

2.5 Transaction Costs and Robustness

A strategy's performance before trading costs may differ from its performance after costs. Frequent position changes can make transaction costs more important. Bessembinder and Chan (1998), for example, show that the interpretation of technical-rule performance depends partly on trading costs. Their cost estimates are specific to their data and setting and should not be transferred directly to this study.

This study examines transaction-cost scenarios of 0, 10, and 25 basis points per one-way position change. These are modeled assumptions for sensitivity analysis, not estimates of the actual trading costs faced by every investor or market participant. The cost-adjusted results should be interpreted accordingly.

The study also compares strategy and buy-and-hold performance by year and examines stock-year observations in which buy and hold has a negative return. Such comparisons can show whether results vary across the observed periods. However, annual comparisons across a small number of securities and years are descriptive; they do not by themselves establish statistical significance or prove that a strategy will be robust in future periods.

Backtests can also be affected by data snooping when many rules or parameter values are examined and the best result is selected after the fact. Sullivan, Timmermann, and White (1999) demonstrate the importance of accounting for this issue when evaluating technical trading rules. The four moving-average windows in this study are compared as specified; the results should not be presented as proof that the best-performing window will continue to be best outside the sample.

For strategy comparisons to be meaningful, the evaluation period must account for the warm-up required to calculate each moving average. The buy-and-hold benchmark and moving-average strategies are evaluated over the same post-warm-up period. The actual start date used for each comparison is reported in the empirical analysis.

2.6 Scope of the Market Comparison

The study compares Apple, Microsoft, and NVIDIA with Kaspi.kz, Halyk Bank, and Kazatomprom. The second group is described as **Kazakhstan-related** because these companies are associated with Kazakhstan but trade under different listing arrangements. The six securities do not represent the entire U.S. or Kazakhstan equity market.

Differences in the results are therefore differences between these securities and portfolios over the periods examined. The study does not measure national differences in liquidity, regulation, investor participation, or market efficiency. Any discussion of those factors is contextual and is not a finding of the analysis.

Chapter Summary

This chapter reviewed the concepts that inform the empirical analysis: realized risk and return, Modern Portfolio Theory, diversification, Sharpe ratios, moving-average strategies, transaction costs, and backtest robustness. These ideas motivate comparisons of selected securities and equal-weighted portfolios, as well as tests of MA20, MA50, MA100, and MA200 against buy and hold. The literature also shows why results should be interpreted cautiously: strategy performance can depend on the selected rule, sample, trading costs, and evaluation period.

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Chapter 3: Data and Methodology

3.1 Data Collection and Sample Selection

This study conducts a comparative quantitative analysis of three U.S. technology equities and three Kazakhstan-related equities. The U.S. securities are Apple Inc. (AAPL), Microsoft Corporation (MSFT), and NVIDIA Corporation (NVDA). The Kazakhstan-related securities are Kaspi.kz (KSPI), Halyk Bank (HSBK.IL), and Kazatomprom (KAP.IL). The latter group includes securities associated with Kazakhstan and is not treated as a representative sample of the entire Kazakhstan stock market.

Historical price data were downloaded from Yahoo Finance using the yfinance Python library. The requested download periods are shown below. The end date supplied to the data request is exclusive.

Group	Symbol	Requested start date	Requested end date
U.S. technology	AAPL	1 January 2020	1 January 2025
U.S. technology	MSFT	1 January 2020	1 January 2025
U.S. technology	NVDA	1 January 2020	1 January 2025
Kazakhstan-related	KSPI	19 January 2024	1 January 2025
Kazakhstan-related	HSBK.IL	1 January 2020	1 January 2025
Kazakhstan-related	KAP.IL	1 January 2020	1 January 2025

The available data periods differ across securities, particularly because the KSPI series used in this study begins in 2024. For the portfolio comparison, returns were aligned by date and observations with missing values were removed. The resulting matched sample covers **22 January 2024 to 31 December 2024**, with **235 observations** for each portfolio.

The six securities have different listing arrangements and trading calendars. Therefore, “U.S. technology equities” and “Kazakhstan-related equities” refer to the selected securities in this study, not to complete national-market portfolios.

3.2 Data Processing and Daily Returns

The analysis uses the Close price field returned by Yahoo Finance. Daily returns are calculated as the percentage change between consecutive available closing prices:

$$r_{i,t} = \frac{P_{i,t}}{P_{i,t-1}} - 1$$

where $P_{i,t}$ is the closing price of security i on date t , and $r_{i,t}$ is its daily return.

Daily returns allow securities with different price levels to be compared on a common percentage basis. They are used to calculate return and risk measures, construct portfolios, estimate correlations, and evaluate moving-average strategies.

Returns calculated from closing prices represent price returns unless the downloaded price series is adjusted to include distributions.

3.3 Return and Risk Measures

The mean daily return for security i is:

$$\bar{r}_i = \frac{1}{T} \sum_{t=1}^T r_{i,t}$$

Daily volatility is measured using the sample standard deviation of daily returns:

$$s_i = \sqrt{\frac{1}{T-1} \sum_{t=1}^T (r_{i,t} - \bar{r}_i)^2}$$

Annualized arithmetic return and annualized volatility are calculated using 252 trading days per year:

$$R_{i,\text{annualized}} = 252\bar{r}_i$$

Annualized arithmetic return is not the same as the compound annual growth rate (CAGR). Cumulative return is calculated by compounding daily returns:

$$R_{i,\text{cumulative}} = \prod_{t=1}^T (1 + r_{i,t}) - 1$$

Cumulative wealth from an initial investment of one unit is:

$$W_{i,t} = \prod_{s=1}^t (1 + r_{i,s})$$

Drawdown at date t is the percentage decline from the previous wealth peak:

$$D_{i,t} = \frac{W_{i,t}}{\max_{1 \leq s \leq t} W_{i,s}} - 1$$

Maximum drawdown is the lowest drawdown observed during the evaluation period:

$$\text{MDD}_i = \min_t (D_{i,t})$$

A 30-day rolling annualized volatility measure is also calculated to examine how portfolio volatility changes over time:

$$\sigma_{30,t} = \sqrt{252} s(r_{t-29}, \dots, r_t)$$

3.4 Risk-Adjusted Performance

Risk-adjusted performance is measured using the annualized Sharpe ratio. The study applies a fixed annual risk-free-rate assumption of 4.97%, converted to a daily rate as follows:

$$r_{f,daily} = (1 + 0.0497)^{1/252} - 1$$

The annualized Sharpe ratio is:

$$\text{Sharpe} = \sqrt{252} \frac{\text{mean}(r_t - r_{f,daily})}{\text{std}(r_t - r_{f,daily})}$$

A higher Sharpe ratio indicates a higher average excess return per unit of return volatility over the evaluation period. The same risk-free-rate assumption is applied to the portfolio and strategy calculations for consistency.

3.5 Correlation and Diversification Measures

Pearson correlation coefficients are calculated between the daily returns of each pair of securities within each group. Correlation ranges from -1 to +1: positive values indicate returns that tend to move together, while negative values indicate returns that tend to move in opposite directions.

To examine changes in correlations over time, 30-day rolling pairwise correlations are calculated. For a group of three securities, the average pairwise rolling correlation on date t is:

$$\bar{\rho}_{30,t} = \frac{1}{3} \sum_{i < j} \rho_{ij,30,t}$$

The mean 30-day rolling correlation is the average of this series over the evaluation period.

The diversification ratio is calculated as:

$$DR = \frac{\sum_{i=1}^N w_i \sigma_i}{\sigma_p}$$

where w_i is the weight of security i , σ_i is its volatility, and σ_p is the volatility of the portfolio. A higher ratio indicates a greater reduction in portfolio volatility relative to the weighted average volatility of its component securities. In this study, the ratio is calculated for equal-weighted portfolios; it is not used to optimize portfolio weights.

3.6 Portfolio Construction

Two equal-weighted portfolios are constructed: a U.S. technology portfolio and a Kazakhstan-related portfolio. Each security receives a weight of one-third:

$$w_i = \frac{1}{3}, \quad i = 1, 2, 3$$

The daily return of portfolio p is:

$$r_{p,t} = \sum_{i=1}^3 w_i r_{i,t}$$

Portfolio return is therefore the arithmetic average of the three component returns on each date. This calculation assumes the portfolio is rebalanced to equal weights daily. Portfolio returns are compared using annualized arithmetic return, annualized volatility, Sharpe ratio, cumulative return, maximum drawdown, rolling volatility, correlation, and diversification ratio.

Portfolio comparisons use the 235-observation matched sample. Any interpretation is limited to the selected securities and this sample period.

3.7 Moving-Average Strategy Design

Four simple moving-average windows are evaluated: MA20, MA50, MA100, and MA200. The simple moving average for window n is:

$$SMA_{n,t} = \frac{1}{n} \sum_{k=0}^{n-1} P_{t-k}$$

The signal is defined as:

$$Signal_{n,t} = \begin{cases} 1, & P_t > SMA_{n,t} \\ 0, & P_t \leq SMA_{n,t} \end{cases}$$

The position applied to returns is the previous day's signal:

$$Position_{n,t} = Signal_{n,t-1}$$

Gross strategy return is calculated as:

$$r_{n,t}^{strategy} = Position_{n,t} r_t$$

The strategy is long the security when the lagged signal is 1 and holds no position when it is 0. The strategy is compared with a buy-and-hold benchmark over the same evaluation period.

Each moving average requires a warm-up period before a valid signal can be calculated. For a fair comparison across the four windows, the evaluation period begins only after MA200 has produced a usable signal. Buy and hold and all four strategies are then evaluated over the same post-warm-up period for each security. Because the signal is lagged by one day, the first strategy return is earned on the trading day after the first valid MA200 signal.

The backtest uses daily closing-price returns and a lagged closing-price signal. It does not model next-day opening prices or intraday execution. Its results should therefore be interpreted as a simplified backtest rather than a record of executable trades.

3.8 Transaction-Cost Analysis

Transaction costs are evaluated under three scenarios: 0, 10, and 25 basis points per one-way position change. One basis point is equal to 0.01%. Turnover on date t is:

$$Turnover_t = |Position_t - Position_{t-1}|$$

Net strategy return is calculated as:

$$r_{n,t}^{net} = Position_{n,t}r_t - \left(\frac{Cost_{bps}}{10,000} \right) Turnover_{n,t}$$

The resulting net returns are used to calculate cumulative return, annualized arithmetic return, annualized volatility, Sharpe ratio, maximum drawdown, and position changes. These cost levels are scenario assumptions; they are not claims about the actual fees faced by every investor or security. The model also does not include taxes, bid–ask spreads beyond the assumed cost, market impact, or a separate terminal liquidation cost.

3.9 Annual and Downside Comparisons

Strategy performance is also summarized by calendar year. The buy-and-hold and strategy returns for year y are compounded from the daily observations in that year:

$$R_y^{strategy} = \prod_{t \in y} (1 + r_t^{strategy}) - 1$$

Annual outperformance is:

$$Outperformance_y = R_y^{strategy} - R_y^{BH}$$

For the downside comparison, stock-year observations with a negative buy-and-hold return are selected:

$$R_y^{BH} < 0$$

The difference between the strategy and buy-and-hold return in those observations is:

$$Difference_y = R_y^{strategy} - R_y^{BH}$$

A positive value means that the strategy returned more than buy and hold for that stock-year. This measure is a return difference; it is not a direct measure of maximum drawdown reduction. Annual and downside comparisons are descriptive summaries of the observed sample.

3.10 Software and Implementation

All data processing and analyses are conducted in Python using Jupyter Notebook. The yfinance library is used to retrieve price data; pandas is used to organize time series and calculate returns, portfolios, and correlations; NumPy is used for numerical calculations; and Matplotlib is used to create figures.

The notebook allows calculations, tables, and visualizations to be examined together.

Chapter 4: Analysis of Selected U.S. Technology Equities

4.1 Scope of the Analysis

This chapter examines Apple (AAPL), Microsoft (MSFT), and NVIDIA (NVDA), the three U.S. technology equities selected for this study. It describes their historical closing prices and daily returns, summarizes their return and volatility characteristics, and reports their pairwise correlations.

The individual-stock analysis uses price data from January 2020 to December 2024. Each series contains 1,258 closing-price observations and 1,257 daily-return observations. Moving-average strategy results are evaluated over a common valid period beginning after the MA200 warm-up period. The matched-sample comparison between the U.S. technology and Kazakhstan-related portfolios is presented in Chapter 6.

4.2 Individual Stock Performance

Table 4.1 reports cumulative return, annualized arithmetic return, annualized volatility, Sharpe ratio, and maximum drawdown for each stock. The results use the common post-warm-up period.

Table 4.1. Individual U.S. Stock Performance, 15 October 2020–31 December 2024

Stock	Cumulative return	Annualized arithmetic return	Annualized volatility	Sharpe ratio	Maximum drawdown
Apple	111.68%	21.46%	26.90%	0.617	−30.91%
Microsoft	97.95%	19.64%	26.05%	0.568	−37.15%
NVIDIA	855.43%	67.20%	52.20%	1.194	−66.34%

NVIDIA recorded the highest cumulative and annualized arithmetic returns among the three stocks during this period. It also had the highest annualized volatility and the largest maximum drawdown. Apple and Microsoft had lower annualized volatility than NVIDIA. Apple had a higher Sharpe ratio than Microsoft, while NVIDIA had the highest Sharpe ratio of the three.

These figures describe realized results for the selected sample period. They do not establish that NVIDIA will continue to have higher returns or risk in the future.

Price and Daily Return Patterns

All three stocks ended the sample at higher price levels than at the beginning, but their paths differed.

AAPL fell during the early part of 2020, then rose through 2020 and 2021. Its price declined during 2022, recovered through much of 2023, and rose again during 2024, ending the sample near USD 250.

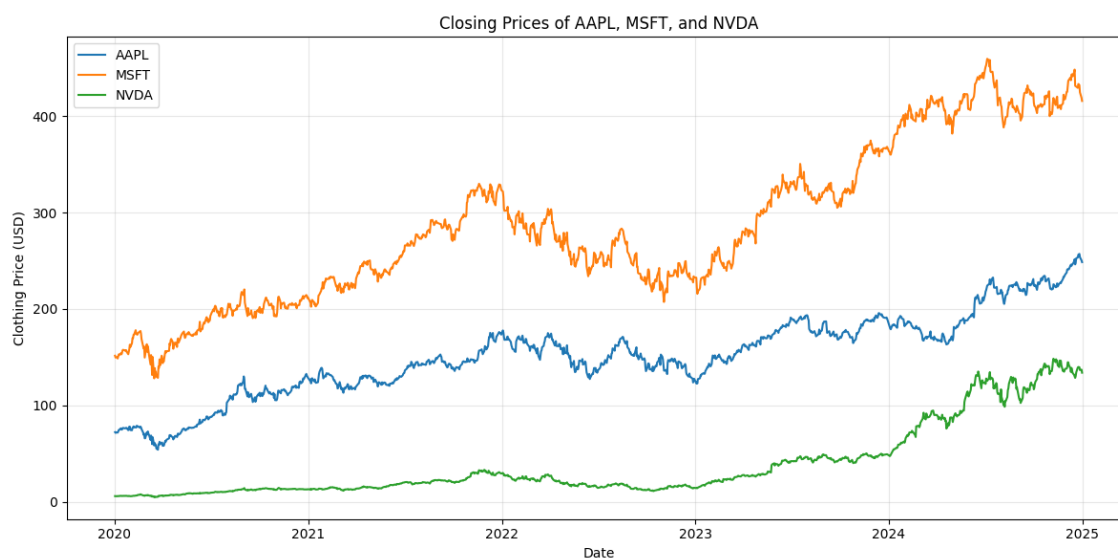


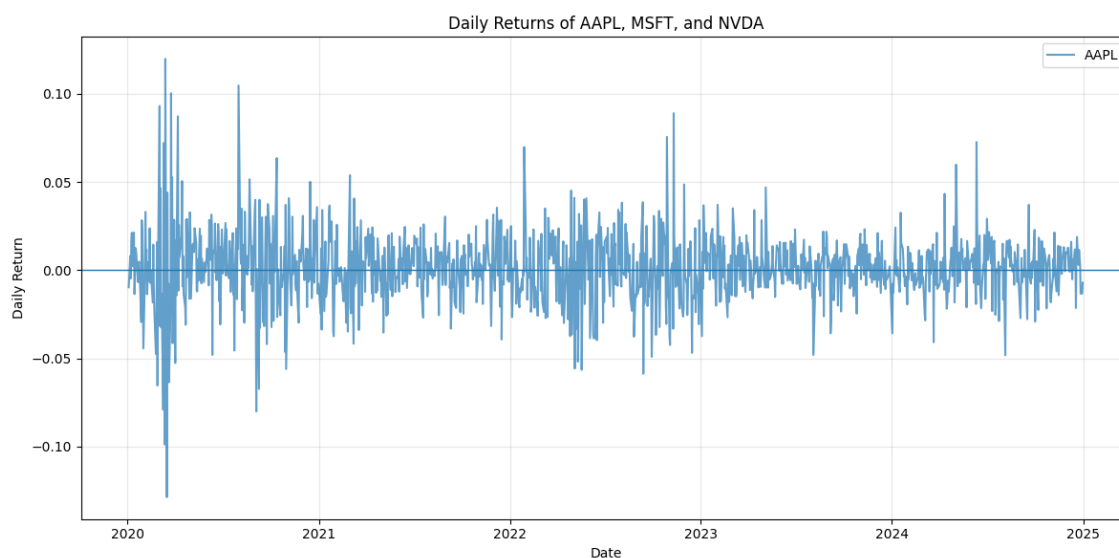
Figure 4.1. Closing Prices of AAPL, MSFT, and NVDA

MSFT also declined in early 2020 before rising through 2020 and 2021. It reached a high near the end of 2021, then fell during 2022. The price rose again through 2023 and into 2024, reaching above USD 400 and ending the sample near USD 420.

NVDA's closing price rose substantially during 2020 and 2021, then declined during 2022. It recovered through 2023 and rose sharply during 2024, with noticeable fluctuations, ending the sample near USD 140.

The figure uses a shared price scale. Since the stocks began at very different price levels, it illustrates their closing-price paths but does not provide a direct comparison of percentage performance. Cumulative wealth or rebased prices are more suitable for comparing relative investment growth.

The daily return plots show short-term changes around each stock's longer-term price path. The histograms show how daily returns are distributed within the sample. NVIDIA's higher volatility in Table 4.1 is consistent with its more widely dispersed daily returns. These descriptive plots do not by themselves explain the causes of large daily movements.



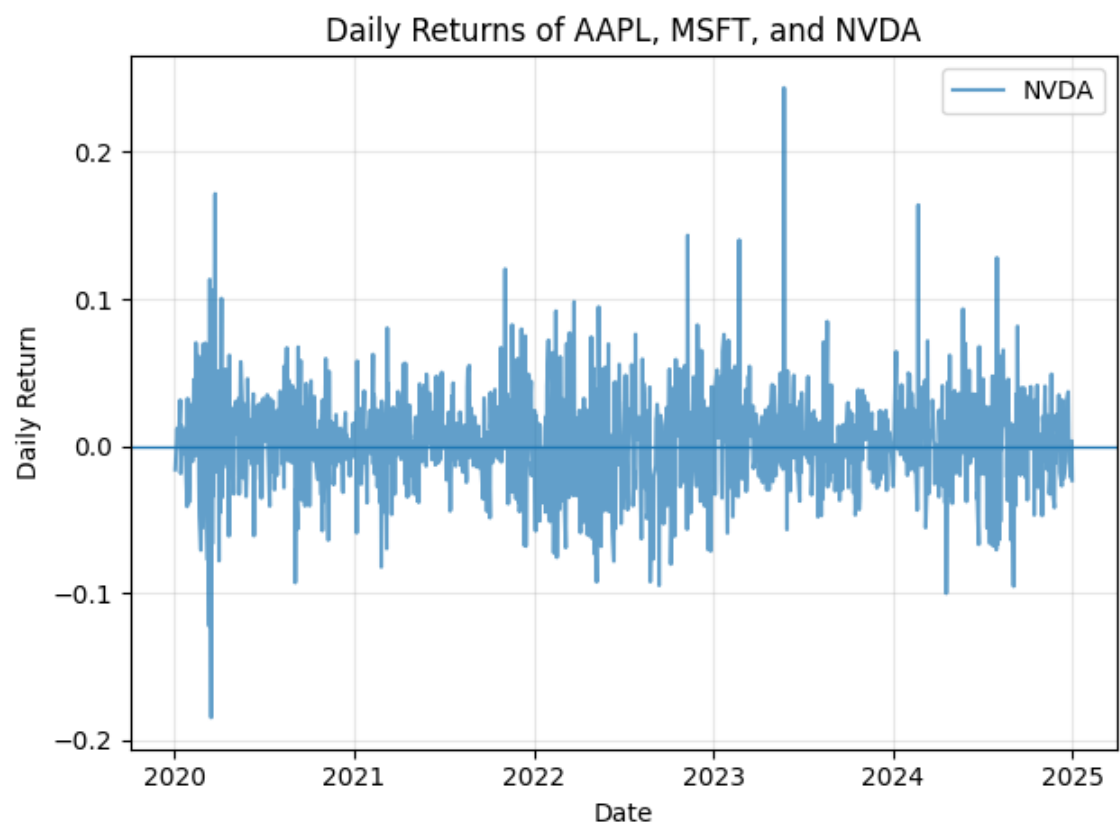
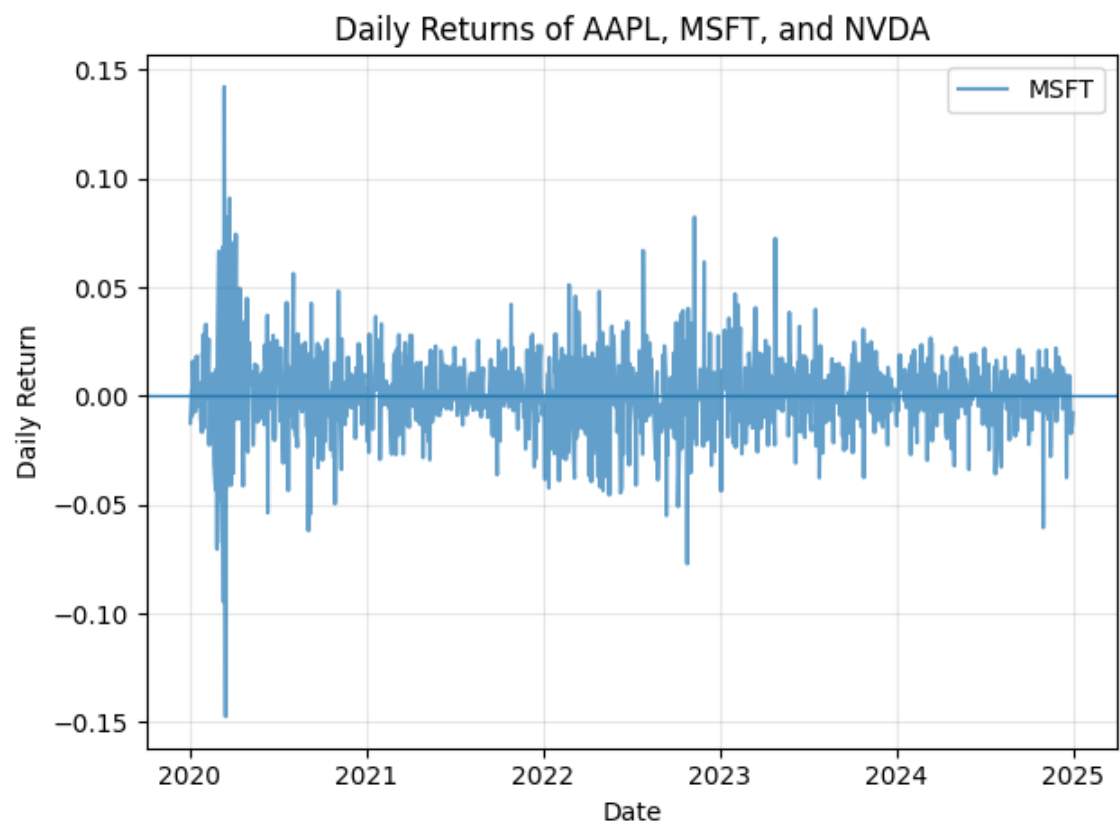


Figure4.2. Daily Returns of AAPL, MSFT, and NVDA

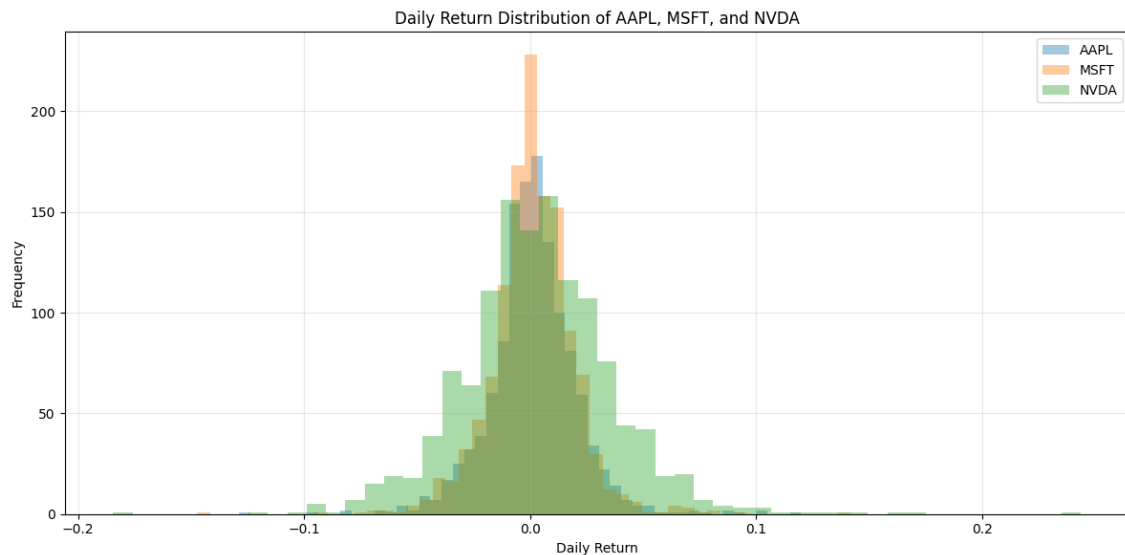


Figure 4.3. Daily Return Distributions of AAPL, MSFT, and NVDA

The daily return plots show short-term changes around each stock’s longer-term price path. The histograms show how daily returns are distributed within the sample. NVIDIA’s higher volatility in Table 4.1 is consistent with its more widely dispersed daily returns. These descriptive plots do not by themselves explain the causes of large daily movement.

4.4 Correlation Analysis

Pairwise Pearson correlations are calculated using the 235-observation matched sample from 22 January to 31 December 2024.

Table 4.2. Correlations Among U.S. Technology Stock Returns

Pair	Correlation
AAPL–MSFT	0.474
AAPL–NVDA	0.251
MSFT–NVDA	0.447

All three correlations are positive in this sample. The AAPL–MSFT pair has the highest correlation, while AAPL–NVDA has the lowest. The lower AAPL–NVDA correlation may contribute to diversification when the securities are combined, although this table alone does not measure each stock’s marginal contribution to portfolio risk reduction. The mean of the three pairwise correlations is approximately **0.391**. These estimates apply only to the selected securities and matched sample period.

4.5 U.S. Equal-Weighted Portfolio

The U.S. technology portfolio assigns an equal weight of one-third to AAPL, MSFT, and NVDA. Its results are calculated over the matched sample of 235 observations.

Table 4.3. U.S. Equal-Weighted Portfolio Performance, Matched Sample

Measure	Result
Mean daily return	0.186%
Daily volatility	1.595%
Annualized arithmetic return	46.86%
Annualized volatility	25.32%
Formal Sharpe ratio	1.659
Cumulative return	50.21%
Maximum drawdown	−16.99%
Diversification ratio	1.266

The portfolio recorded a cumulative return of 50.21% over the matched sample. Its annualized arithmetic return was 46.86%, with annualized volatility of 25.32%. The formal Sharpe ratio, calculated using the risk-free-rate assumption specified in Chapter 3, was 1.659. The portfolio’s maximum drawdown during the sample was −16.99%.

The diversification ratio was greater than one, indicating a reduction in volatility relative to the weighted average volatility of the component stocks. This result describes the equal-weighted portfolio during the matched sample; it does not show that equal weighting is an optimal allocation.

Figure 4.4. Cumulative Wealth of the U.S. Equal-Weighted Portfolio

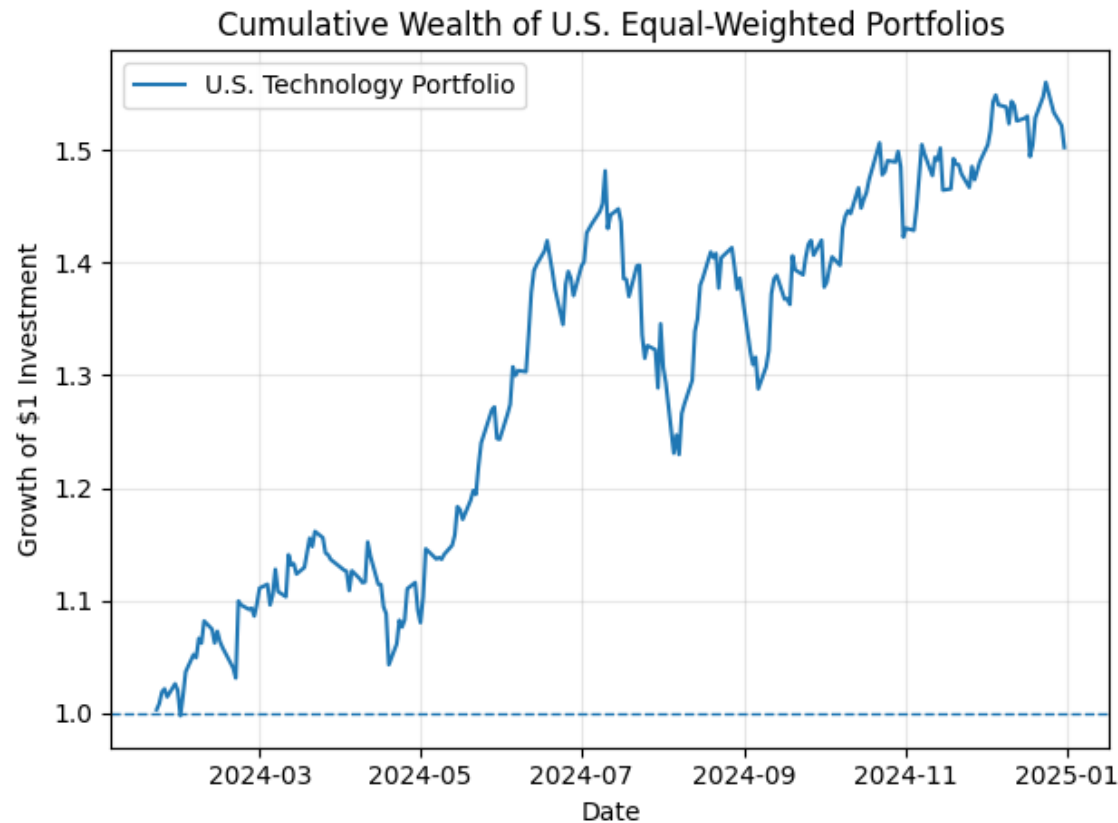
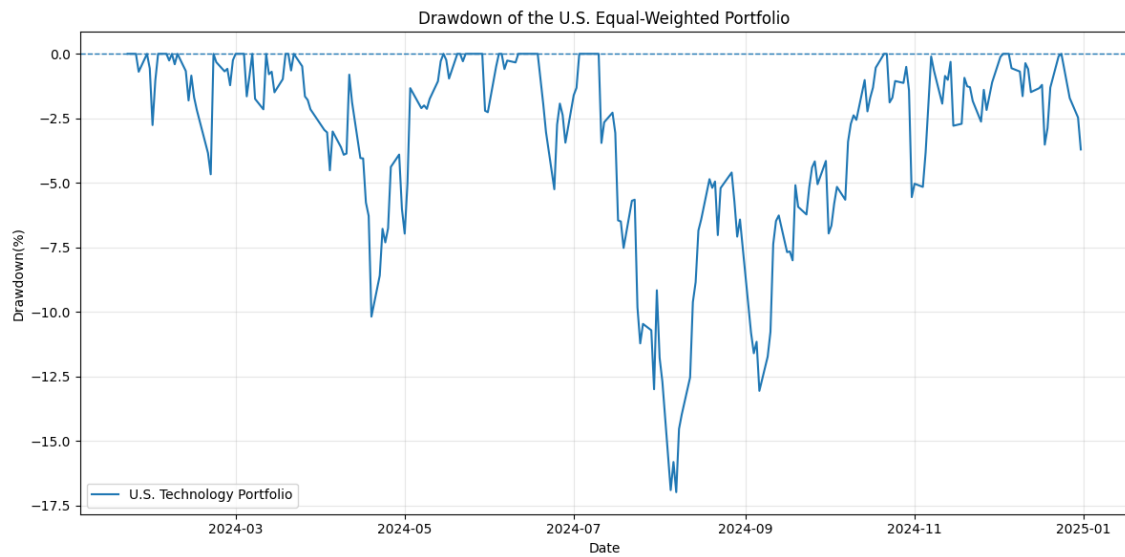


Figure 4.5. Drawdown of the U.S. Equal-Weighted Portfolio



4.6 Moving-Average Strategy Results

The moving-average strategies hold a stock when its closing price is above its moving average and otherwise hold no position. Signals are shifted by one trading day. Table 4.4 compares buy and hold with MA20, MA50, MA100, and MA200 over each stock's common post-MA200 evaluation period. The reported returns do not include transaction costs; transaction-cost scenarios are presented separately.

Table 4.4. Moving-Average Strategy Performance, Common Post-Warm-Up Period

Stock	Strategy	Cumulative return	Annualized arithmetic return	Annualized volatility	Sharpe ratio	Maximum drawdown
Apple	Buy & Hold	111.68%	21.46%	26.90%	0.617	-30.91%
Apple	MA20	117.88%	20.25%	18.56%	0.830	-19.54%
Apple	MA50	56.01%	12.30%	18.56%	0.402	-29.27%
Apple	MA100	42.84%	10.48%	19.98%	0.282	-26.33%
Apple	MA200	67.95%	14.69%	21.68%	0.454	-31.19%
Micro soft	Buy & Hold	97.95%	19.64%	26.05%	0.568	-37.15%
Micro soft	MA20	7.12%	3.23%	17.80%	-0.091	-40.93%
Micro soft	MA50	17.91%	5.64%	18.55%	0.042	-22.89%
Micro soft	MA100	36.95%	9.25%	18.79%	0.234	-22.24%

Micro soft	MA200	59.19%	12.99%	19.62%	0.415	−30.49%
NVID IA	Buy & Hold	855.43%	67.20%	52.20%	1.194	−66.34%
NVID IA	MA20	522.82%	50.56%	37.83%	1.208	−33.86%
NVID IA	MA50	434.55%	48.02%	40.73%	1.060	−32.00%
NVID IA	MA100	513.95%	51.68%	41.54%	1.127	−46.37%
NVID IA	MA200	947.16%	65.52%	44.18%	1.373	−40.99%

For Apple, MA20 produced a higher Sharpe ratio and lower annualized volatility than buy and hold. Its annualized arithmetic return was slightly lower, while its cumulative return was slightly higher over the evaluation period.

For Microsoft, all four moving-average strategies had lower annualized arithmetic returns and lower Sharpe ratios than buy and hold. MA50, MA100, and MA200 had smaller maximum drawdowns than buy and hold, while MA20 had a larger maximum drawdown.

For NVIDIA, MA20 and MA200 had Sharpe ratios above that of buy and hold. MA200 also recorded a higher cumulative return, although its annualized arithmetic return was slightly lower. All four moving-average strategies had lower annualized volatility and smaller maximum drawdowns than buy and hold in the saved results.

The results therefore differ by stock and performance measure. A strategy with a higher Sharpe ratio does not necessarily have a higher annualized arithmetic return, and these historical results do not establish that the same strategy will perform similarly in other periods.

4.7 Chapter Summary

This chapter examined three selected U.S. technology stocks and their equal-weighted portfolio. NVIDIA had the highest return, volatility, and maximum drawdown among the individual stocks over the common post-warm-up period. In the 2024 matched sample, the equal-weighted U.S. portfolio had a cumulative return of 50.21%, an annualized volatility of 25.32%, and a formal Sharpe ratio of 1.659.

The moving-average results varied by stock and window. MA20 had a higher Sharpe ratio than buy and hold for Apple, while MA20 and MA200 had higher Sharpe ratios for NVIDIA. None of the tested moving averages had a higher Sharpe ratio than buy and hold for Microsoft. These findings are specific to the observed periods and do not imply universal strategy effectiveness.

Chapter 5: Kazakhstan-Related Equity Analysis

This chapter examines the historical performance and portfolio characteristics of three Kazakhstan-related securities: Kaspi.kz (KSPI), Halyk Bank (HSBK.IL), and Kazatomprom (KAP.IL). It describes their price movements, daily returns, and return distributions, then evaluates an equal-weighted portfolio of the three securities.

The securities have different listing histories and available data periods. Their price and return plots use the available data for each security, while the descriptive statistics and portfolios comparisons uses the common matched sample where indicated. The portfolio analysis uses the dates on which all three securities have observations. Cross-group comparisons with the U.S. technology portfolio are presented in Chapter 6.

5.1 Descriptive Statistics

Table 5.1 presents descriptive statistics for the three securities over the common matched sample from 22 January 2024 to 31 December 2024, consisting of 235 daily-return observations for each security. Annualized arithmetic return and annualized volatility are calculated from daily returns using 252 trading days. The Sharpe ratio uses the study's common annual risk-free rate of 4.97%.

Table 5.1. Descriptive Statistics of Kazakhstan-Related Equities

Security	Sample period	Observations	Mean daily return	Annualized arithmetic return	Annualized volatility	Sharpe ratio
KSPI	2024-01-22—2024-12-31	235	0.000526	13.25%	0.400147	0.211108
HSBK.IL	2024-01-22—2024-12-31	235	0.001232	31.06%	0.432082	0.606495
KAP.IL	2024-01-22—2024-12-31	235	0.001508	37.99%	0.456791	0.725521

Among the three securities, KAP.IL had the highest annualized arithmetic return and volatility in its available sample. HSBK.IL had the next-highest figures for both measures. KSPI had the lowest annualized arithmetic return and volatility, as well as the lowest Sharpe ratio.

These figures describe realized historical returns and volatility over the same matched period and are not forecasts. Using a common sample ensures that differences across the three securities are not driven by different observation periods.

5.2 Price Behavior

Figure 5.1 plots the closing prices of KSPI, HSBK.IL, and KAP.IL over their available sample period

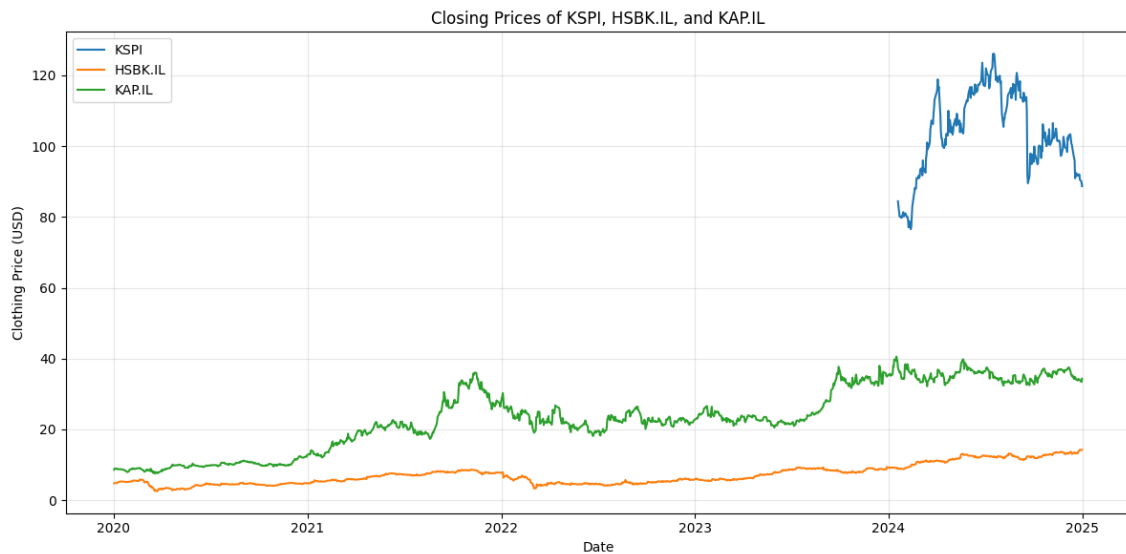


Figure 5.1. Closing Prices of KSPI, HSBK.IL, and KAP.IL

KAP.IL rose from around USD 9 in 2020 to above USD 30 in 2021, then declined during 2022. Its price moved at lower levels through much of 2022 and 2023 before rising again during 2024. HSBK.IL declined in early 2020 and then followed a general upward path, reaching approximately USD 14 by the end of 2024. KSPI appears only from 2024 in this dataset: its closing price rose above USD 120 during the year before falling to around USD 89 at year-end.

The chart uses one vertical scale for all three securities, even though their price levels differ substantially. The movements of HSBK.IL are consequently less visible.

5.3 Daily Return Analysis

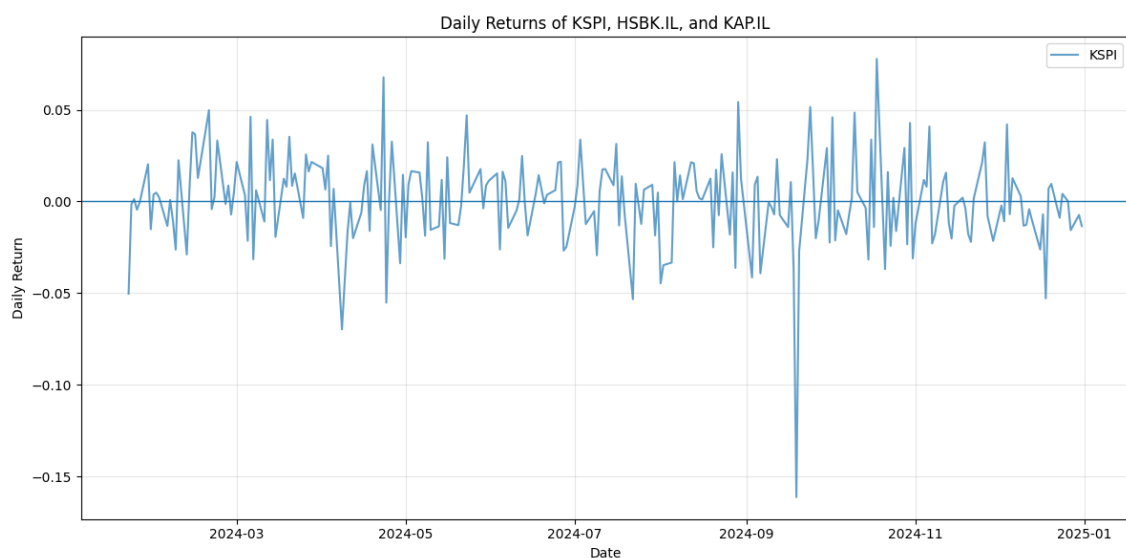
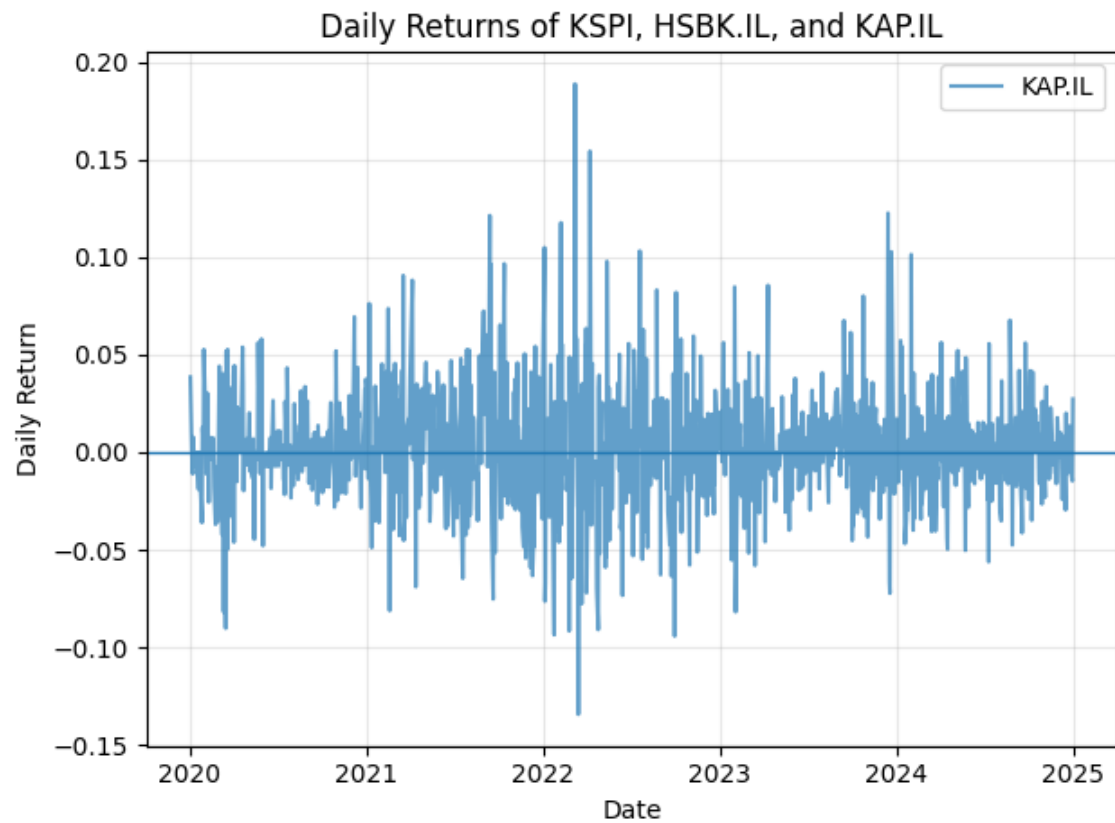
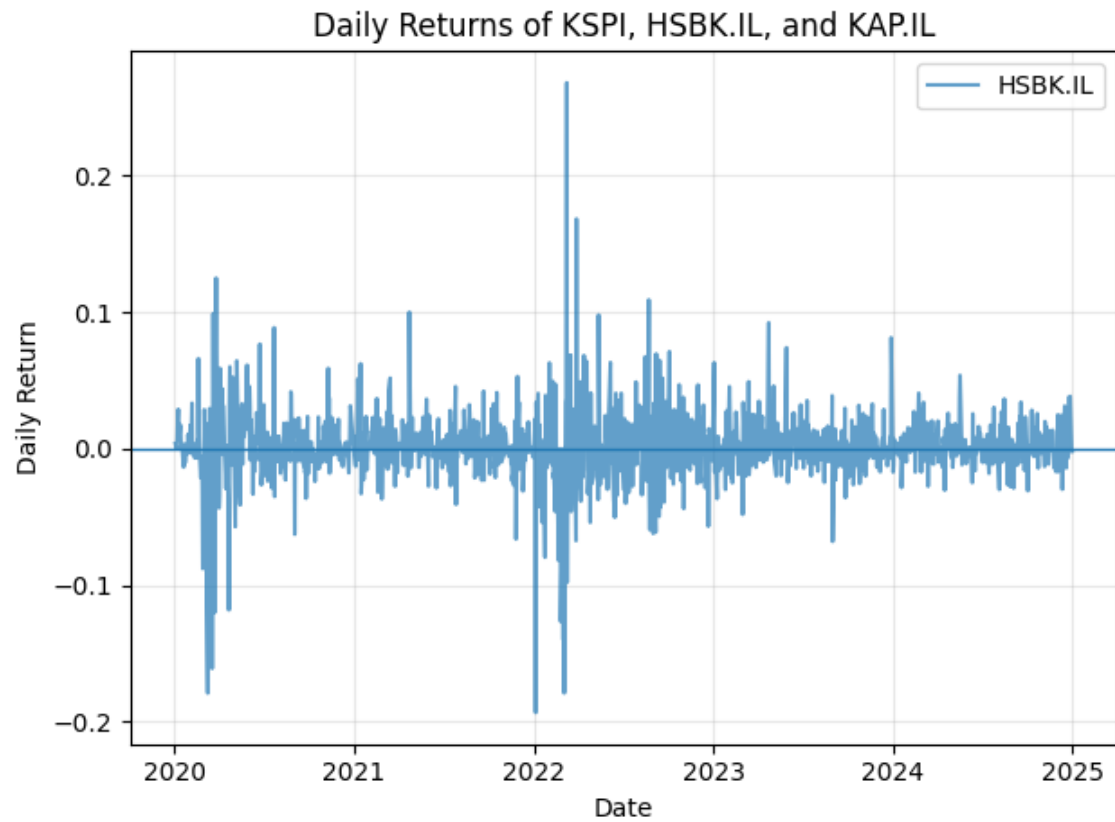


Figure 5.2. Daily Returns of KSPI, HSBK.IL, and KAP.IL

Figure 5.2 presents daily returns for the three securities. Daily returns make it easier to compare short-term price changes than closing prices, whose scales may differ.



Daily returns fluctuate around zero for all three securities, with occasional larger movements in both directions. The KSPI series contains a particularly large negative daily return around September 2024. HSBK.IL shows several pronounced movements, especially around 2020 and 2022. KAP.IL also displays large return movements, with several notable fluctuations around 2022. These plots show the timing and scale of observed daily changes; they do not identify the events that caused them.

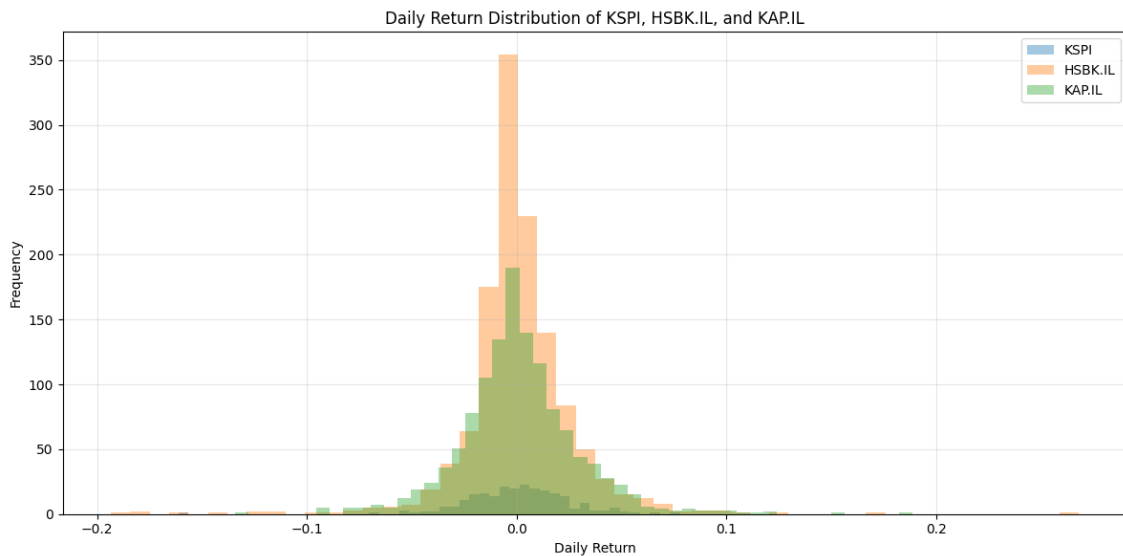


Figure 5.3. Daily Return Distributions of KSPI, HSBK.IL, and KAP.IL

The three distributions are concentrated around zero, but each contains observations farther from the center. HSBK.IL and KAP.IL show wider visible ranges of daily returns than KSPI in this plot. However, the chart displays raw frequencies, and the sample sizes differ: the notebook reports 239 observations for KSPI and 1,261 for each of HSBK.IL and KAP.IL. Therefore, the histogram bar heights should not be used to compare the securities directly. A density histogram would make the distributions easier to compare.

5.4 Correlation and Diversification

Table 5.2 reports pairwise correlations calculated over the 235-observation matched sample from 22 January 2024 to 31 December 2024.

Table 5.2. Pairwise Correlations Among Kazakhstan-Related Equities

	KSP I	HSBK.I L	KAP.I L
KSPI	1.00 0	0.100	0.086
HSBK.I L	0.10 0	1.000	0.160

KAP.IL	0.08 6	0.160	1.000
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The pairwise correlations were positive but relatively low in this matched sample. The highest correlation was between HSBK.IL and KAP.IL (0.160), while the lowest was between KSPI and KAP.IL (0.086). The average of the three pairwise correlations was 0.116.

These results describe how the three securities' returns moved together during the matched period. They do not establish that the correlations will remain at these levels in other periods.

5.5 Equal-Weighted Portfolio Performance

An equal-weighted portfolio was constructed with one-third of its weight assigned to each security. Its performance was evaluated over the same matched sample used in the correlation analysis.

Table 5.3. Kazakhstan-Related Equal-Weighted Portfolio Performance

Measure	Value
Sample period	22 Jan 2024–31 Dec 2024
Observations	235
Annualized arithmetic return	19.95%
Annualized volatility	20.99%
Sharpe ratio	0.720
Maximum drawdown	−12.32%
Cumulative return	18.01%
Diversification ratio	1.541

Figure 5.4 shows the cumulative wealth of a hypothetical USD 1 investment in the portfolio.

Over the matched sample, the portfolio ended with approximately USD 1.18 for each initial USD 1 invested, equivalent to a cumulative return of 18.01%. The path included periods of both growth and decline; the line alone does not show how large those declines were relative to earlier peaks.

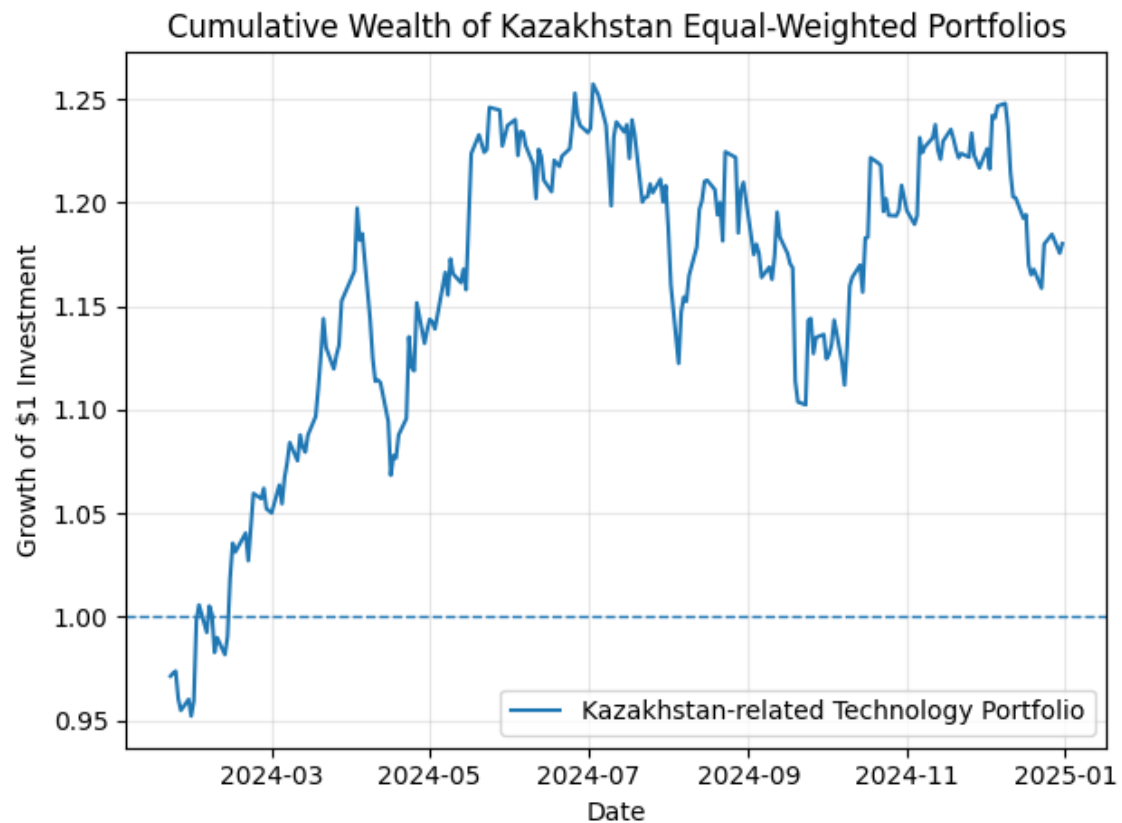


Figure 5.4. Cumulative Wealth of the Kazakhstan-Related Equal-Weighted Portfolio

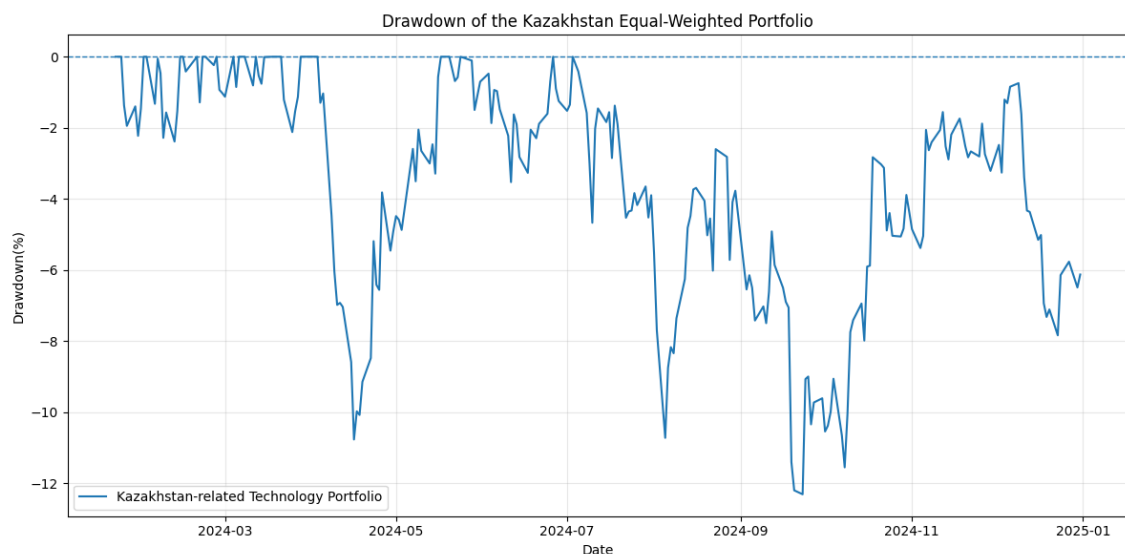


Figure 5.5. Drawdown of the Kazakhstan-Related Equal-Weighted Portfolio

The portfolio's maximum drawdown was approximately 12.32%, reached on 23 September 2024. This measure describes the largest decline from a previous portfolio peak during the matched sample. The diversification ratio was 1.541, indicating volatility reduction under this equal-weight construction. The portfolio uses fixed equal weights and was not optimized to maximize returns or minimize risk.

5.6 Moving-Average Strategy Results

The notebook evaluates Buy & Hold and MA20, MA50, MA100, and MA200 strategies over a common valid period for each security, beginning after the MA200 warm-up period. Table 5.4 summarizes the Sharpe ratios reported for those periods. The Sharpe ratio uses the study's annual risk-free rate of 4.97%.

Table 5.4. Sharpe Ratios for Buy & Hold and Moving-Average Strategies

Security	Evaluation start	Buy & Hold	MA20	MA50	MA100	MA200
KSPI	1 Nov 2024	-2.925	-1.836	-5.715	Not meaningful ¹	-3.336
HSBK.IL	14 Oct 2020	0.799	0.505	0.569	0.993	1.110
KAP.IL	14 Oct 2020	0.757	-0.057	0.329	0.113	0.233

For HSBK.IL, MA100 and MA200 had higher Sharpe ratios than Buy & Hold over the evaluated period. For KAP.IL, all four moving-average strategies had lower Sharpe ratios than Buy & Hold. KSPI's valid evaluation period begins only on 1 November 2024, leaving roughly two months of data; its strategy results are too limited for a reliable conclusion.

KSPI's MA100 strategy had zero return volatility in this short evaluation period, so the Sharpe ratio is undefined. The notebook output reports it as negative infinity; that value should not be interpreted as a meaningful estimate.

These are gross strategy results and do not account for the transaction-cost scenarios. The strategy periods also differ between securities, so the table supports within-security comparisons only.

5.7 Chapter Summary

This chapter examined the price and return behavior of KSPI, HSBK.IL, and KAP.IL. In their available individual samples, KAP.IL had the highest annualized arithmetic return and volatility, while KSPI had the lowest Sharpe ratio. Over the 235-observation matched sample, the equal-weighted portfolio had an annualized arithmetic return of 19.95%, annualized volatility of 20.99%, a Sharpe ratio of 0.720, and a maximum drawdown of 12.32%.

The matched-sample pairwise correlations were low and positive. Moving-average results varied by security: some HSBK.IL strategies had higher Sharpe ratios than Buy & Hold, while none of the tested KAP.IL moving averages did. KSPI's short valid strategy period limits interpretation. These findings apply to the selected securities and sample periods, not to the Kazakhstan stock market as a whole. Chapter 6 compares the Kazakhstan-related and U.S. technology portfolios over their matched sample.

Chapter 6: Comparative Analysis

This chapter compares the selected U.S. technology equities—AAPL, MSFT, and NVDA—with the Kazakhstan-related equities—KSPI, HSBK.IL, and KAP.IL. To make the portfolio comparison consistent, both equal-weighted portfolios are evaluated over the matched period from 22 January 2024 to 31 December 2024, with 235 observations.

The comparison concerns these selected securities and this sample period. It should not be interpreted as a comparison of the entire U.S. and Kazakhstan stock markets.

6.1 Matched-Sample Portfolio Performance

Table 6.1 compares the two equal-weighted portfolios using annualized arithmetic return, annualized volatility, Sharpe ratio, maximum drawdown, cumulative return, and diversification ratio.

Table 6.1. Matched-Sample Performance of Equal-Weighted Portfolios

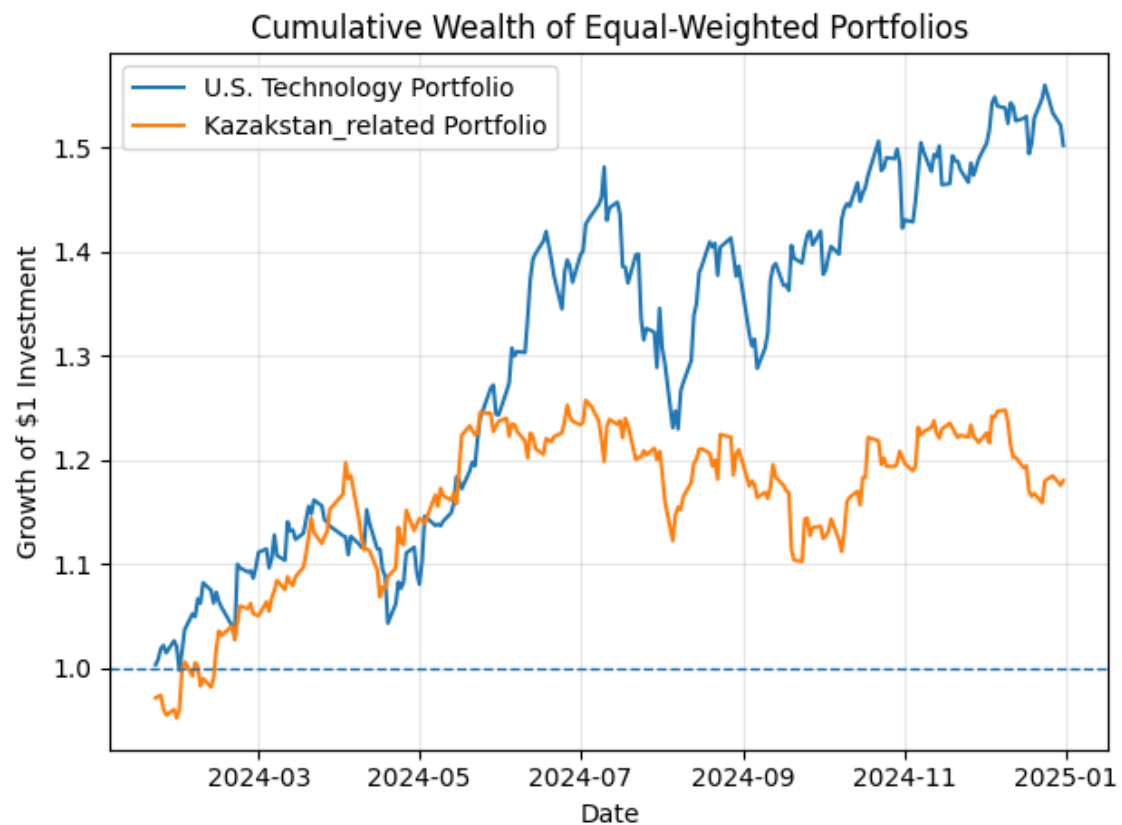
Measure	U.S. Technology Portfolio	Kazakhstan-Related Portfolio
Sample period	22 Jan 2024–31 Dec 2024	22 Jan 2024–31 Dec 2024
Observations	235	235
Annualized arithmetic return	46.86%	19.95%
Annualized volatility	25.32%	20.99%
Sharpe ratio	1.659	0.720
Maximum drawdown	−16.99%	−12.32%
Cumulative return	50.21%	18.01%
Diversification ratio	1.266	1.541

Over this matched period, the U.S. technology portfolio had higher annualized arithmetic and cumulative returns. It also had higher annualized volatility and a larger maximum drawdown. Its Sharpe ratio was higher, indicating stronger risk-adjusted performance according to this measure during the sample.

The Kazakhstan-related portfolio had lower volatility and a smaller maximum drawdown over the same period. It also had a higher diversification ratio. These results describe the two equal-weighted portfolios in this sample; they do not establish that one portfolio would have the same relative performance in another period.

6.2 Cumulative Wealth

Figure 6.1 shows the growth of a hypothetical USD 1 invested in each equal-weighted portfolio over the matched sample.



By the end of the sample, the U.S. portfolio had grown to approximately USD 1.50, while the Kazakhstan-related portfolio had grown to approximately USD 1.18. The paths also show interim fluctuations, so the ending values should be read alongside the volatility and drawdown results in Table 6.1.

Figure 6.1. Cumulative Wealth of Equal-Weighted Portfolios

6.3 Risk and Drawdown

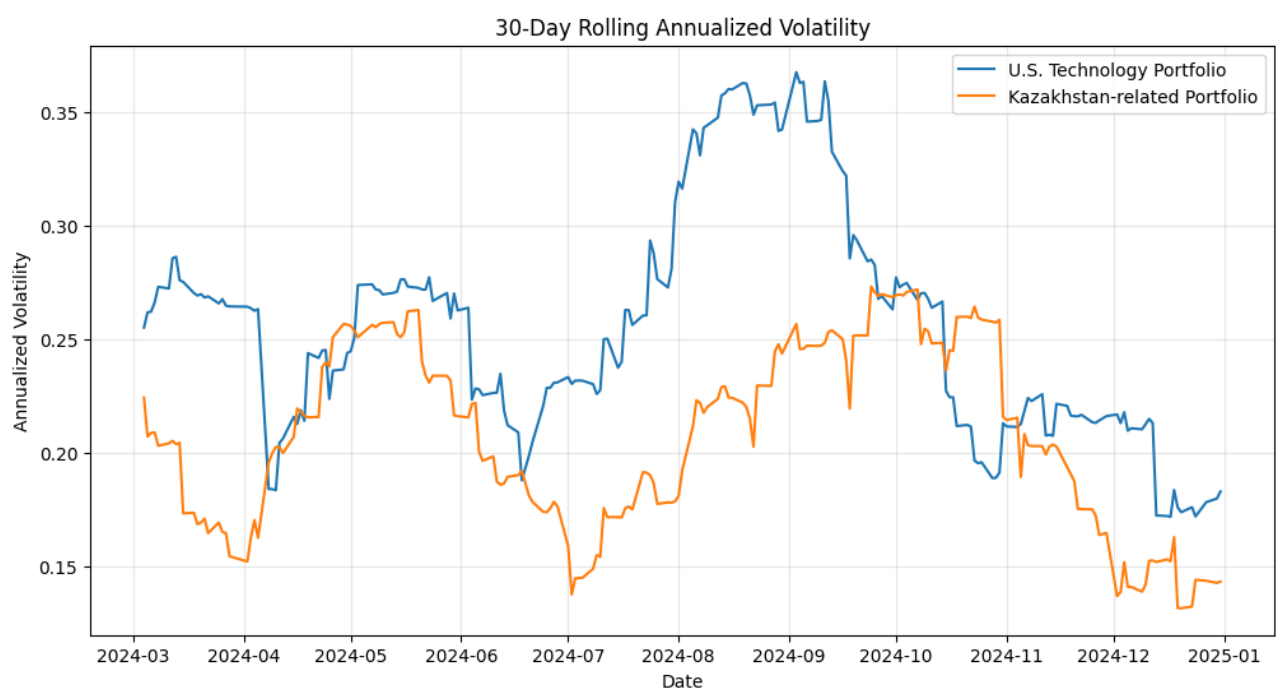


Figure 6.2 compares the 30-day rolling annualized volatility of the portfolios. The average rolling volatility was approximately 25.54% for the U.S. technology portfolio and 20.77% for the Kazakhstan-related portfolio. In this sample, the U.S. portfolio’s rolling volatility reached a higher maximum, while the Kazakhstan-related portfolio’s rolling volatility was lower on average.

Figure 6.3 compares portfolio drawdowns. The U.S. portfolio’s maximum drawdown was approximately 16.99%, reached on 7 August 2024. The Kazakhstan-related portfolio’s maximum drawdown was approximately 12.32%, reached on 23 September 2024.

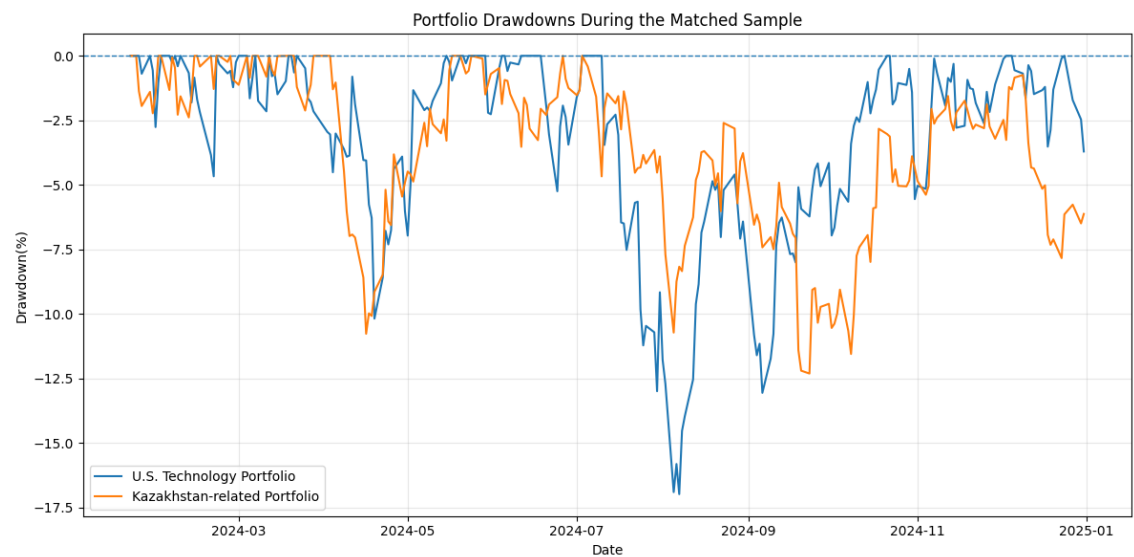


Figure 6.3. Portfolio Drawdowns During the Matched Sample

The drawdown results show that the U.S. portfolio experienced the deeper peak-to-trough loss during the matched period. This complements the volatility comparison: volatility summarizes the variability of returns, while maximum drawdown measures the largest observed decline from a previous wealth peak.

6.4 Correlation and Diversification

Table 6.2 presents the pairwise correlations among the securities in each group, calculated over the matched sample.

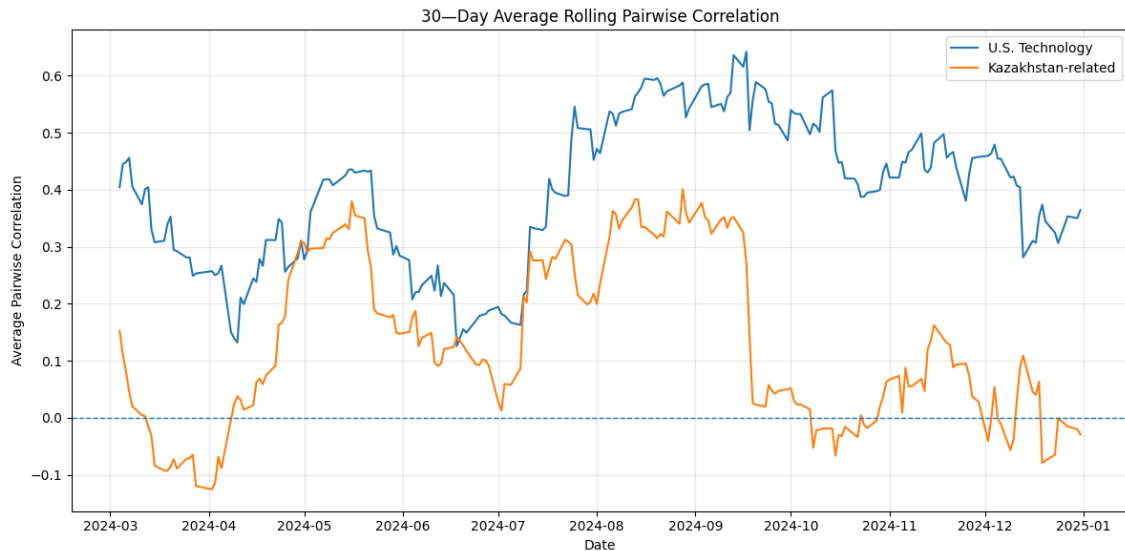
Table 6.2. Pairwise Correlations Within Each Equity Group

U.S. technology pair	Correlation	Kazakhstan-related pair	Correlation
AAPL–MSFT	0.474	KSPI–HSBK.IL	0.100
AAPL–NVDA	0.251	KSPI–KAP.IL	0.086
MSFT–NVDA	0.447	HSBK.IL–KAP.IL	0.160
Average	0.391	Average	0.116

The average pairwise correlation was lower among the Kazakhstan-related securities than among the U.S. technology securities during the matched sample. The diversification ratios were 1.541 and 1.266, respectively. Together, these results are consistent with greater volatility reduction for the equal-weighted Kazakhstan-related portfolio in this sample.

Figure 6.4 shows how average pairwise rolling correlations changed over time.

Figure 6.4. 30-Day Average Rolling Pairwise Correlation



The rolling correlations varied over the sample rather than remaining constant. The U.S. group's average rolling correlation was 0.397, with a range of approximately 0.126 to 0.642. The Kazakhstan-related group's average was 0.129, with a range of approximately -0.126 to 0.401. These values come from the notebook's separate calculation of all three unique pairs. The final summary function currently omits one pair, so its rolling-correlation values should not be used.

6.5 Moving-Average Strategy Comparison

The moving-average strategies were compared with Buy & Hold for each security over a common valid period beginning after the MA200 warm-up. The results differ by security and moving-average window; there is no single strategy that outperformed Buy & Hold across all six securities.

For the U.S. securities, MA20 had a higher Sharpe ratio than Buy & Hold for AAPL, while MA200 had a higher Sharpe ratio for NVDA. None of the four moving-average strategies had a higher Sharpe ratio than Buy & Hold for MSFT. For the Kazakhstan-related securities, MA100 and MA200 had higher Sharpe ratios than Buy & Hold for HSBK.IL, while none of the tested windows did so for KAP.IL. KSPI's valid strategy period begins on 1 November 2024, leaving only a short evaluation period, so its strategy results are not reliable enough for a strong comparison.

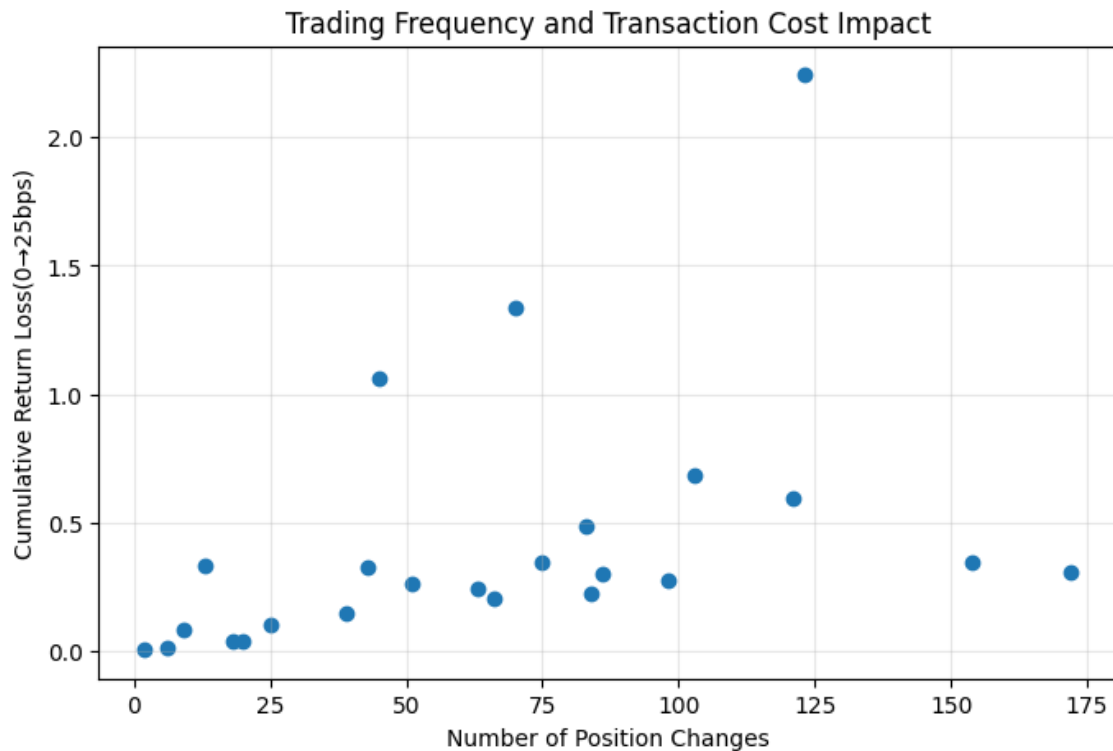
These results should be interpreted within each security because the evaluation periods differ. They also describe gross strategy results before transaction costs. The cost analysis is reported separately.

6.6 Transaction-Cost Sensitivity

The notebook evaluates one-way transaction-cost scenarios of 0, 10, and 25 basis points. In the reported backtests, cumulative returns declined as the assumed transaction cost increased. For example, AAPL's MA20 cumulative return declined from 117.88% at 0 basis points to 98.19% at 10 basis points and 71.90% at 25 basis points.

Figure 6.5 plots the number of position changes against the cumulative return loss between the 0- and 25-basis-point scenarios.

Figure 6.5. Trading Frequency and Transaction-Cost Impact



The notebook reports a correlation of approximately 0.492 between position changes and return loss across the tested stock-strategy combinations. This indicates a positive association in this set of backtests: combinations with more position changes tended to show larger cost-related return losses. It is a descriptive relationship and does not by itself show that trading frequency caused the difference.

6.7 Temporal and Downside Robustness

The notebook also includes annual outperformance and downside-year analyses. However, the current annual-results function calculates the MA200 start date but does not apply that date before calculating yearly returns. As a result, the current yearly outperformance chart and summary may include warm-up-period dates as zero strategy returns.

However, the current annual-results function calculates the MA200 start date but does not apply that date before calculating yearly returns. The annual and downside robustness results should be added to this section only after the function is corrected and the results are recalculated. Until then, the

figures should not be treated as evidence that the strategy conclusions are robust across years or during negative Buy & Hold years.

6.8 Chapter Summary

Over the matched period from 22 January to 31 December 2024, the U.S. technology portfolio had higher annualized arithmetic return, cumulative return, and Sharpe ratio than the Kazakhstan-related portfolio. It also had higher annualized volatility and a larger maximum drawdown. The Kazakhstan-related portfolio had a higher diversification ratio and lower average pairwise correlation within its group over this sample.

Moving-average results varied across securities and windows. Some strategies had higher Sharpe ratios than Buy & Hold for AAPL, NVDA, and HSBK.IL, while no tested window did so for MSFT or KAP.IL. KSPI's short valid evaluation period limits interpretation. Transaction-cost scenarios reduced strategy returns, with greater position-change counts associated with larger return losses in the tested combinations.

These are sample-specific findings for six selected securities and the stated periods. The annual robustness analysis remains to be recalculated before it can support a conclusion about performance stability across years or downside periods.

Chapter 7: Conclusion

7.1 Summary of the Study

This study compared the historical return, risk, and diversification characteristics of three U.S. technology equities—Apple (AAPL), Microsoft (MSFT), and NVIDIA (NVDA)—with three Kazakhstan-related equities: Kaspi.kz (KSPI), Halyk Bank (HSBK.IL), and Kazatomprom (KAP.IL). It also examined equal-weighted portfolios and tested moving-average trend-following strategies against a Buy & Hold benchmark.

For the portfolio comparison, both portfolios were evaluated over the matched period from 22 January to 31 December 2024, covering 235 observations. This matched period helps ensure that the portfolio results refer to the same dates. The individual-security and strategy analyses use their available valid periods, which differ by security.

7.2 Main Findings

Over the matched sample, the U.S. technology portfolio had an annualized arithmetic return of 46.86%, compared with 19.95% for the Kazakhstan-related portfolio. Its cumulative return was also higher: 50.21%, compared with 18.01%. The U.S. portfolio had higher annualized volatility (25.32% versus 20.99%) and a larger maximum drawdown (−16.99% versus −12.32%). Its Sharpe ratio was 1.659, compared with 0.720 for the Kazakhstan-related portfolio.

The Kazakhstan-related portfolio had a higher diversification ratio (1.541 versus 1.266). The average pairwise correlation in the matched sample was also lower among the Kazakhstan-related securities (0.116) than among the U.S. technology securities (0.391). Within this sample, these results are consistent with stronger volatility reduction from combining the Kazakhstan-related securities at equal weights. They do not show that the same relationship would hold in other periods.

Moving-average results varied across securities and windows. Some moving-average strategies had higher Sharpe ratios than Buy & Hold for AAPL, NVDA, and HSBK.IL, while none did for MSFT or KAP.IL. KSPI's valid evaluation period after the MA200 warm-up was very short, so its strategy results are not conclusive. The tested transaction-cost scenarios reduced cumulative strategy returns, and strategies with more position changes tended to have larger cost-related return losses in this set of backtests.

7.3 Economic Interpretation and Future Outlook

The empirical results provide a basis for interpreting the differences between the two groups of equities. However, these differences should not be attributed to national market characteristics alone. The U.S. sample consists of three technology companies—Apple, Microsoft, and NVIDIA—whereas the Kazakhstan-related sample includes a fintech company, a bank, and a natural-resources company. The comparison therefore reflects differences in both geography and sector composition.

During the matched sample, the U.S. technology portfolio recorded a higher annualized arithmetic return than the Kazakhstan-related portfolio, but it also had higher annualized volatility and a larger maximum drawdown. This pattern illustrates a return–risk trade-off within the observed period. Innovation-related areas such as artificial intelligence, semiconductors, cloud computing, and

digital services may provide useful context for interpreting the U.S. companies' performance. However, this study did not test the effects of these factors, interest rates, corporate investment, or market valuations on stock returns. They should therefore be treated as possible contextual explanations, rather than causes established by the analysis.

The three Kazakhstan-related securities have different business exposures. Kaspi.kz is associated with digital financial and consumer services, Halyk Bank with banking, and Kazatomprom with uranium and natural resources. Their different exposures may help explain why their returns had relatively low pairwise correlations in the matched sample. The Kazakhstan-related portfolio also had a higher diversification ratio than the U.S. technology portfolio. These results suggest that sectoral differences within the selected group may have contributed to its diversification characteristics. The analysis does not, however, establish the cause of the correlation differences or show that the same pattern applies to Kazakhstan's equity market as a whole.

This distinction matters because all three U.S. securities in the study belong to the technology sector, while the Kazakhstan-related securities span several sectors. The portfolio comparison therefore cannot isolate a country effect from a sector effect or the characteristics of individual companies. The lower correlations observed within the Kazakhstan-related group should be understood as a result for these three securities during the matched sample, rather than as a general feature of the national market.

Future performance may be influenced by factors such as technology investment, semiconductor demand, monetary conditions, developments in digital financial services, and conditions in uranium and other commodity markets. These factors may also change the relationships among the securities over time. The historical results in this study are not forecasts, and the analysis does not determine how those factors will affect future returns.

Future research could help separate country, sector, and company-specific effects by using broader samples, comparing sector-matched portfolios, including market indices, and extending the common sample period. Such research could test whether the differences observed here persist across more securities and market conditions.

7.4 Limitations

Several limitations affect how these results should be interpreted.

First, the study covers only six selected securities. The three Kazakhstan-related securities do not represent the entire Kazakhstan stock market, and the three U.S. technology companies do not represent the U.S. equity market. Conclusions should therefore be limited to the securities and periods examined.

Second, the matched portfolio sample covers less than one year. Although the portfolios are compared over the same dates, this short period cannot establish whether their relative performance persists over longer market cycles.

Third, the securities have different data histories. KSPI's available history is shorter, and its valid MA200 strategy period covers only a small part of 2024. Its strategy results are especially sensitive to the short evaluation period.

Fourth, the backtest uses historical closing prices and simplified moving-average signals. Actual trading may occur at different prices and may involve slippage, taxes, liquidity constraints, and other costs. The transaction-cost scenarios in this study do not represent every possible real-world trading cost.

7.5 Future Research

Future research could extend the matched period as more KSPI data become available and include a broader set of securities from each group. It could also compare the selected securities with appropriate market indices, examine alternative portfolio weighting methods, and test additional transaction-cost and execution assumptions.

Future research could evaluate moving-average strategies over longer common samples and examine the performance across different market conditions. Future work could add year-by-year and downside-period comparisons over longer samples. Such work would help determine whether the observed strategy results remain similar across different periods and assumptions.

7.6 Final Conclusion

This study found that, over the matched 2024 sample, the U.S. technology portfolio achieved higher return and Sharpe ratio, while the Kazakhstan-related portfolio had lower volatility, a smaller maximum drawdown, and a higher diversification ratio. Moving-average performance was mixed across securities, and transaction costs reduced strategy returns.

These findings provide a descriptive comparison of six selected equities and two equal-weighted portfolios. They should not be generalized to the full U.S. or Kazakhstan stock markets, or interpreted as evidence that the strategies will perform similarly in the future.